SECTION–I(i) : (Maximum Marks : 24)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

1. Manganese salt + PbO_2 + conc. $HNO_3 \longrightarrow$ the solution acquires purple colour. The colour is due to :-
(A) $HMnO_4$ (B) $Pb(NO_3)_2$ (C) $Mn(NO_3)_2$ (D) MnO_2

2. When borax is heated on platinum loop wire forms a glass like bead which is made-up of:-
(A) Sodium tetraborate
(B) Sodium metaborate
(C) Sodium metaborate and boric anhydride
(D) Boric anhydride and sodium tetraborate

3. A light green coloured salt soluble in water gives black precipitate on reaction with Na_2S, which dissolves readily in HCl. the metal ion present is:-
(A) Co^{2+} (B) Fe^{2+} (C) Ni^{2+} (D) Mn^{2+}

4. Which metal gives blue ash when its salt is heated with Na_2CO_3 solid and $Co(NO_3)_2$ on a charcoal piece?
(A) Cu (B) Mg (C) Al (D) Zn

5. Cl_2 gas can **NOT** be prepared by –
(A) Electrolysis of fused NaCl
(B) Acidification of Bleaching powder
(C) Reaction of conc. HCl with $KMnO_{4(aq.)}$
(D) Heating KCl with conc. H_2SO_4

6. Acidic aqueous solution contain $CuSO_4$, $ZnSO_4$, $MgSO_4$, $CdSO_4$ when react with $H_2S_{(g)}$, Ions which is/are precepitated as metal sulphide.
(A) Cu^{+2}, Cd^{+2} (B) only Cu^{+2} (C) Mg^{+2}, Zn^{+2} (D) Cu^{+2}, Zn^{+2}, Cd^{+2}

7. Select **CORRECT** match
(A) Ag_2CrO_4 – Yellow ppt. (B) $[Fe(SCN)_3]$ – Green ppt.
(C) CdS – Black ppt. (D) $3HgO.HgCO_3$ – Reddish brown ppt.

8. Which of the following salt is **NOT** soluble in HNO_3 (1 : 1) but soluble in aquaregia?
(A) PbS (B) Ag_2S (C) HgS (D) Ag_2CO_3

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SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –2 In all other cases.

9. Which of the following cation(s) will give precipitate with reagent NH_4Cl and NH_4OH :
(A) Ca^{2+} (B) Al^{3+} (C) Bi^{3+} (D) Mg^{2+}

10. Which of the following reagent will give test with Fe^{3+} ions?
(A) NH_4SCN (B) $K_3[Fe(CN)_6]$ (C) NH_4OH (D) $K_4[Fe(CN)_6]$

11. Which of the following metal cation(s) gives white ppt. with Na_2CO_3?
(A) Fe^{3+} (B) Mg^{2+} (C) Cr^{3+} (D) Zn^{2+}

12. Select reaction in which precipitate is formed?
(A) $Ag^+ + SO^{-2}_{3(excess)} \longrightarrow$ (B) $Ba^{2}_{(aq.)} + S_2O_3^{2-} \longrightarrow$
(C) $Cu^{2}_{(aq.)} + S_2O_3^{2-} \longrightarrow$ (D) $Hg^{+2}_{(aq.)} + S^{2-} \longrightarrow$

13. Select the following reaction(s) in which chemical composition of product precipitate is changed when exposed to air for some time :-
(A) $BaCl_2$(aq.) + Na_2SO_3 (aq.) → **ppt - 1**
(B) $FeSO_4$ (aq.) + NaOH (aq.) → **ppt - 2**
(C) BaS + $ZnSO_4$ (aq.) → **ppt - 3**
(D) $MnSO_4$ (aq.) + NaOH (aq.) → **ppt - 4**

14. A test tube contain aqueous solution of $CuSO_4$ which can precipitate metal ion –
(A) $H_2S_{(g)}$ (B) $HCl_{(g)}$ (C) $NH_{3(g)}$ (excess) (D) $PH_{3(g)}$

SECTION–I(iii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in <u>one of the following categories</u> :

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

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Paragraph for Q. No. 15 to 16

In the group analysis of cations, group reagents are used to precipitate out cations.

15. Cations of which group are precipitated in the form of sulphides :-

(A) Group II and III (B) Group II and IV

(C) Group II and V (D) Group VI and III

16. During the analysis of group III cation, NH_4OH is added in presence of NH_4Cl. Which of the following factor has no significance for the addition of NH_4Cl :-

(A) NH_4Cl decreases the concentration of OH^- by common ion effect

(B) NH_4Cl form buffer solution with NH_4OH

(C) Concentration of NH_4^+ ion is decreased by the NH_4Cl addition

(D) All of these

Paragraph for Q. No. 17 to 18

The basic radical are detected by various reagent through precipitation reaction and some of the precipitate soluble in excess of reagent.

17. Which of the following cation does not form precipitate on reaction with KI.

(A) Pb^{+2} (B) Cu^{+2} (C) Bi^{+3} (D) Zn^{+2}

18. Which of the following cation does not form soluble complex with excess of NaOH?

(A) Cr^{+3} (B) Zn^{+2} (C) Cu^{+2} (D) Sn^{+2}

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SECTION–I(iii) : (Maximum Marks : 03)

- This section contains **ONE** questions.
- **Each question has matching lists.** The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : –1 In all other cases

19. Match the folloiwng list :

List - I (Compounds)	**List - II (Reagents)**
(P) $PbCl_2$	(1) Insoluble in water but soluble in NH_3 solution
(Q) AgCl	(2) Insoluble in cold water but soluble in hot water
(R) $FeCl_3$(Neutral solution)	(3) Produces prussian blue on reaction with $K_4[Fe(CN)_6]$
(S) $ZnCl_2$	(4) Soluble in excess of $NaOH_{(aq.)}$

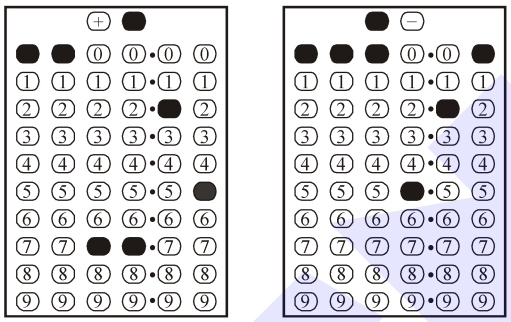
Code :

	(P)	(Q)	(R)	(S)
(A)	2	1	3	4
(B)	2	1	4	3
(C)	3	4	1	2
(D)	1	2	4	3

SECTION-II : (Maximum Marks: 06)

- This section contains **TWO** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +3 If ONLY the correct numerical value is entered as answer.

  Zero Marks : 0 In all other cases.

1. Total number of yellow precipitates which is/are soluble in yellow ammonium sulphide (YAS).
CdS, Bi_2S_3, SnS, SnS_2, As_2S_3, As_2S_5, Sb_2S_3, Sb_2S_5, CuS, HgS

2. How many of the following cations gives ppt with H_2S in acidic solution :-
Zn^{2+} , Cu^{2+} , Ni^{2+} , Sr^{2+} , Al^{3+} , Bi^{3+} , Sn^{2+} , Pb^{2+}